Run report: run-20261011T1813Z-bench

Eigen-Séta v2 Rust KEM: benchmark run (descriptive record)

Study: research/eigen_seta_rust_20261011 · sources: commit 19c641d121097e80f727bcc650a08632e204ba1d

What ran

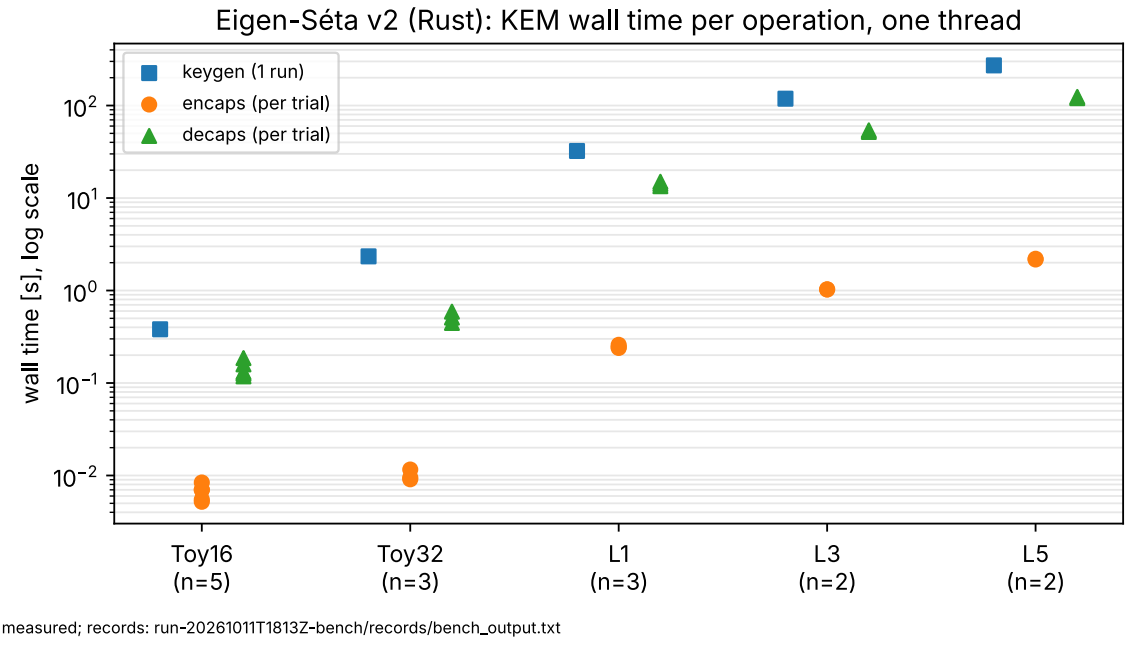
Start (UTC)	2026-10-11T18:13:51Z
Sources	`eigen_seta/` at commit 19c641d12. The binary was built from that tree just before the commit, and a cargo rebuild after the commit was a no-op.
Toolchain	rustc 1.97.0 (2d8144b78 2026-07-07). Release: opt-level 3, thin LTO, 1 codegen unit, default target-cpu.
Host	Intel Xeon Processor @ 2.80 GHz, 4 logical cores, 15 GiB, Linux 6.18.44 x86_64 (adx, avx2, avx512f, bmi2, popcnt)
Isolation	Shared cloud VM. No pinning and no frequency control. One benchmark process at a time.
Command	`bench_field`, then `run Toy16 5`, `run Toy32 3`, `run L1 3`, `run L3 2`, `run L5 2`
Workload	One keygen per set, then the stated number of encaps and decaps trials. Keys and ciphertexts come from OsRng.
Output pin	Each trial checks that decaps returns the encapsulated key. Result: 15/15 correct.
Records	`records/bench_output.txt`, `records/host_and_command.txt`

Requested: 5 sets, 15 trials, and the field micro-benchmark. Completed: all. Failures: none.

Measured values (wall clock, one thread)

Set	keygen (s), n=1	encaps (s), per trial	decaps (s), per trial	correct
Toy16	0.381	0.0055, 0.0070, 0.0070, 0.0084, 0.0052	0.131, 0.159, 0.186, 0.126, 0.118	5/5
Toy32	2.341	0.0095, 0.0116, 0.0091	0.510, 0.446, 0.591	3/3
L1	32.279	0.2448, 0.2588, 0.2394	14.223, 14.997, 13.424	3/3
L3	118.589	1.0256, 1.0304	51.834, 53.940	2/2
L5	271.791	2.2045, 2.1651	123.953, 120.180	2/2

Public key = ciphertext sizes: 216, 376, 1344, 1992 and 2632 bytes, from Toy16 to L5. Secret-degree bit lengths in this run: 51, 80, 278, 409 and 536.



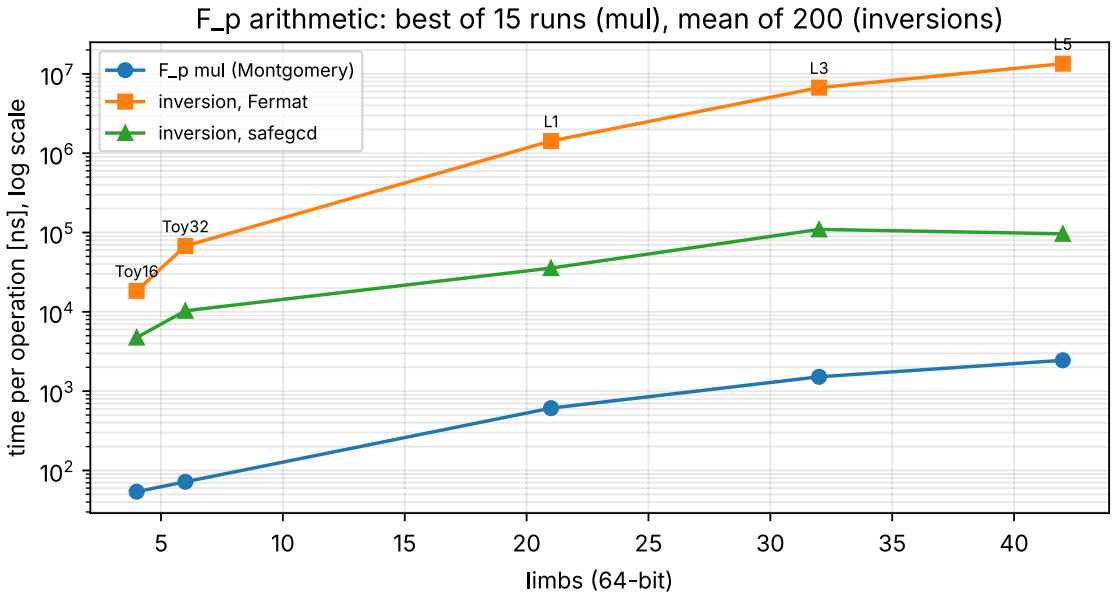
measured; records: run-20261011T1813Z-bench/records/bench_output.txt

Figure 1: Wall time per operation, every trial shown, log scale. Measured; records/bench_output.txt.

Set	limbs	mul (ns), min of 15	inv. Fermat (ns), mean of 200	inv. safegcd (ns), mean of 200
Toy16	4	54	18,403	4,741
Toy32	6	72	67,657	10,316

L1	21	611	1,426,022	35,608
L3	32	1,516	6,708,390	109,506
L5	42	2,449	13,453,052	96,327

No spread is recorded beyond the table. The L3 safegcd figure is above the L5 one. That is what was recorded, and this run does not explain it.



measured; records: run-20261011T1813Z-bench/records/bench_output.txt

Figure 2: F_p multiplication and inversion against limb count, log scale. Measured; records/bench_output.txt.